

A modern chip is a relay race between specialist champions: lithography, materials, process tools, EDA/IP, packaging, and test. Each discipline represents decades of deep expertise, and the strongest semiconductor ecosystems are those that collaborate seamlessly and move at industrial speed. So when Avi Avula, President of Applied Materials India, says “no sovereign silicon,” it is not a critique of India’s Atmanirbhar ambition. Rather, it’s a call to pursue strategic leverage over isolation.

“A real value here is to become a critical player in the global supply chain where you have something to give and have something to get,” Avula says. “So, everybody needs you and you need everybody else.”

If India becomes a place the global chain routes through – because it offers capability, trust, and scale – then India gains resilience and momentum.

Where Applied Materials fits in India’s chip destiny

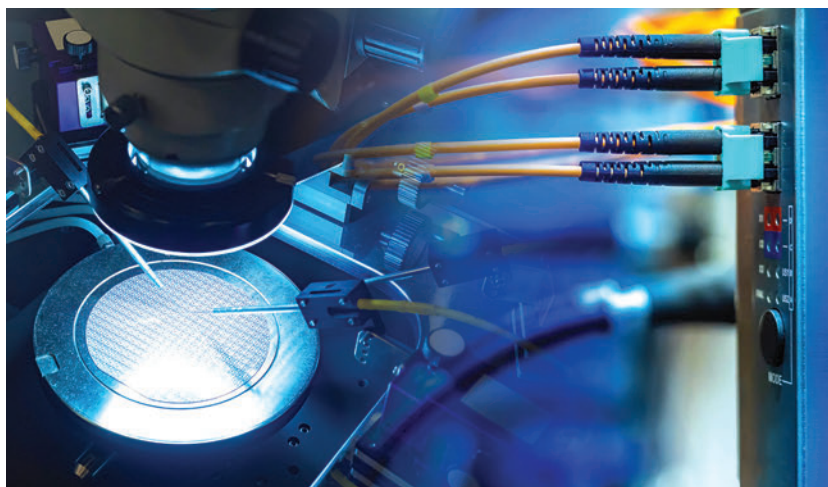
For those unaware, Applied Materials is the “how chips are made” company, responsible for the process technology and manufacturing equipment that sits inside fabs. Avula grounds the conversation in the first thing glossy narratives forget: materials – the foundation on which every advanced chip is built.

“For example, we use a lot of crazy materials here. On a periodic table of elements, we probably use 80% of it. And some of these are obviously very hazardous,” Avula says. “But in California, we have people living – I mean, I lived less than one kilometre from one of these buildings because these are all self-contained in a vacuum environment where there are multiple layers of protection where there is no scenario for leakage.

Applied’s India posture, as Avula tells it, is proof-by-operation. “We are already building things, doing hard things here. We operate India’s

only private-sector facility currently capable of processing 300-millimeter semiconductor wafers,” Avi says, with pride. And he adds why that matters: “It’s a mini-size version of what Tata is building, maybe 1% of the size, but we actually have something that’s operational.”

That “operational” word is everything, if you think about it. In semiconductors, a running line becomes a live classroom – it allows engineers to learn yield, do contamination control, spares logistics, safety procedures, and the unglamorous discipline of keeping a complex process stable across months. It’s where credibility is minted, ultimately.



Zoom out from any one facility and Avula’s diagnosis of India starts to make sense. “We have a rapidly growing end market,” he says. “What is missing is sort of what we’ve called the value chain void as such.” In a summary that couldn’t be any simpler, Avi says, “Design is there, but there are missing pieces.”

His remedy is ecosystem-building by attraction. “Getting a few queen bees is very important, because you can’t pull everything on your own,” he says. The idea is to land “critical players in each value chain segment,” because they “will go and get 50 or 100 other companies to come.” He’s seen how quickly that works when

an anchor commits. “The moment we announced that we are going to be there for a certain sector, then everybody who’s a partner to us or wanted to be a partner to us said, I’m in,” he explains.

India must fix frameworks, codes, infrastructure

For Avula, India doesn’t fall short in ambition. It’s about scaffolding, especially needed for a highly sophisticated, multi-disciplinary exercise as setting up a burgeoning semiconductor ecosystem.

“While India’s working on it, currently there’s no framework for what a semiconductor policy should be

for,” he says. “What kind of chemicals are you allowed to use? How do you use water in this industry? How much power do you need? Where should you zone these things? All these things need to be clearly mentioned,” Avula says.

And it’s for good reason. Because setting up a fab or chemical gas processing plant elsewhere to support the fab is different from building codes that currently apply to factories from other industries, suggests Avi.

“Applied Materials has taken the lead on one dimension to work with a group of companies in our space to write up and create some guidelines for policymakers to look